C11 Introduction to Organic Chemistry – Knowledge Organiser

AQA A-level Chemistry • Set 1: Key Definitions **Paper 2**

#	Question	Answer
1	Hydrocarbon	Compound of hydrogen and carbon only
2	Saturated	Only single C–C bonds
3	Unsaturated	Contains a C=C double bond
4	Homologous series	Same functional group and general formula; each member differs by CH_2
5	Functional group	Group of atoms responsible for a compound's characteristic reactions
6	General formula	Algebraic formula for any member of a homologous series
7	Molecular formula	Actual number of atoms of each element
8	Structural formula	Shows arrangement of atoms, e.g. $CH_3CH_2CH_3$
9	Displayed formula	Shows every atom and every bond
10	Skeletal formula	Carbon skeleton and functional groups only; H on C not shown

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AQA A-level Chemistry • Set 2: Isomerism **Paper 2**

#	Question	Answer
1	Structural isomers	Same molecular formula, different structural formula
2	Three types of structural isomerism	Chain, position, functional group
3	Stereoisomers	Same structural formula, different spatial arrangement
4	Why E–Z isomerism occurs	Restricted rotation about C=C
5	Requirement for E–Z isomerism	Two different groups on each C of the C=C
6	*Z* isomer	Higher-priority groups on the same side
7	*E* isomer	Higher-priority groups on opposite sides
8	CIP priority is based on	Atomic number
9	Functional group isomer of propanal	Propanone
10	Number of structural isomers of C_4H_{10}	2

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AQA A-level Chemistry • Set 3: Naming: Stems and Suffixes **Paper 2**

#	Question	Answer
1	Stem for 1 carbon	meth-
2	Stem for 2 carbons	eth-
3	Stem for 3 carbons	prop-
4	Stem for 4 carbons	but-
5	Stem for 5 carbons	pent-
6	Stem for 6 carbons	hex-
7	Suffix for an alkane	-ane
8	Suffix for an alkene	-ene
9	Suffix for an alcohol	-ol
10	Suffix for an aldehyde	-al

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AQA A-level Chemistry • Set 4: Naming: Functional Groups and Rules **Paper 2**

#	Question	Answer
1	Suffix for a ketone	-one
2	Suffix for a carboxylic acid	-oic acid
3	Prefix for a chlorine substituent	chloro-
4	Prefix for a bromine substituent	bromo-
5	Name of the $-CH_3$ substituent	methyl
6	Prefix for two identical substituents	di-
7	Prefix for three identical substituents	tri-
8	Order of substituent prefixes	Alphabetical
9	Rule for numbering the carbon chain	Lowest possible numbers
10	Prefix for a ring of carbons	cyclo-

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AQA A-level Chemistry • Set 5: Functional Groups and General Formulae **Paper 2**

#	Question	Answer
1	Alcohol functional group	–OH
2	Aldehyde functional group	–CHO
3	Ketone functional group	C=O within the chain
4	Carboxylic acid functional group	–COOH
5	Alkene functional group	C=C
6	Halogenoalkane functional group	C–X
7	General formula of alkanes	C_nH_{2n+2}
8	General formula of alkenes	C_nH_{2n}
9	General formula of cycloalkanes	C_nH_{2n}
10	General formula of alcohols	$C_nH_{2n+1}OH$